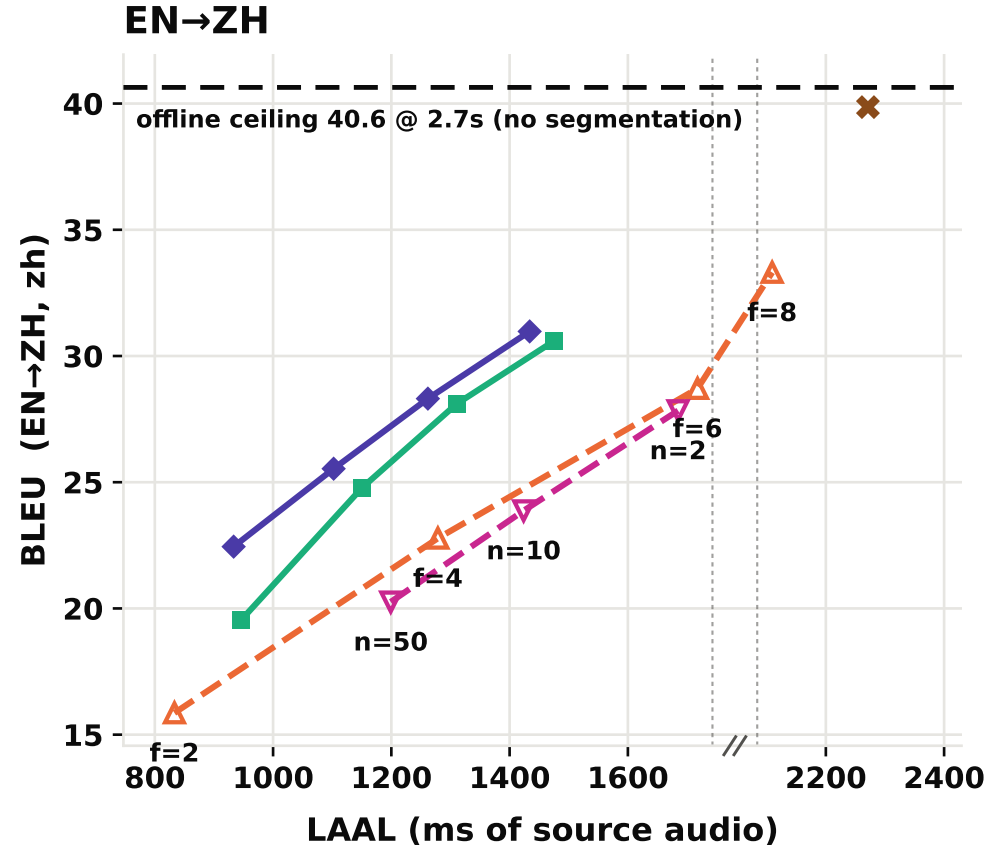
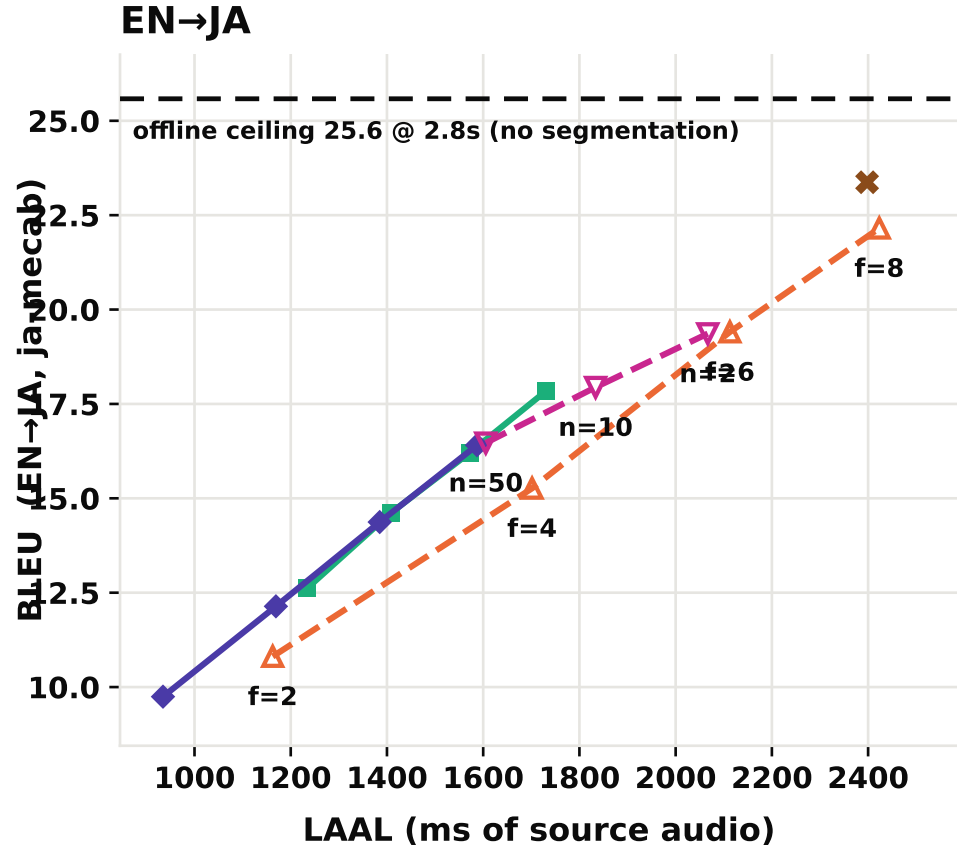
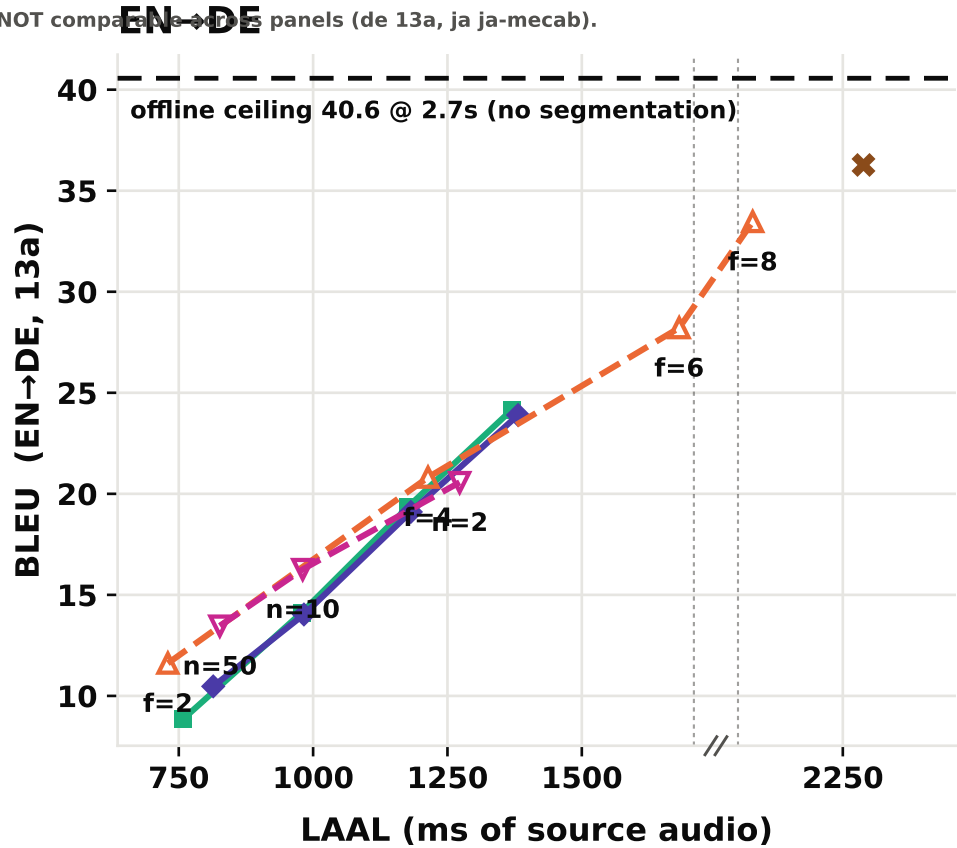


BLEU-latency trade-off on CoVoST2 en->X test 15,530 (madlad) (same translator, madlad400-3b-mt; T = 2/3/4/6 per curve)

Upper-left is better. LAAL is forced-aligned (Qwen3-ForcedAligner; wav2vec2 CTC agrees within 22 ms). Empty x ranges are compressed (break marks on the axis).

Punctuation has no latency knob (it segments below the T budget), so it is one point; it and prefix-match MU sit in a slower band — shown, not matched.

BLEU is NOT comparable across panels (de 13a, ja ja-mecab).



— — Full-sentence offline (ceiling)

—■— Causal align (TransLLaMa; Koshkin et al., 2024)

—◆— SASST (Yang et al., 2026)

—▲— AlignAtt native f-sweep (Papi et al., 2023)

—▽— Prefix-match MU native n-sweep (Zhang et al., 2020)

—x— Punctuation (no latency knob)